□

5.10. Frobenius reciprocity

A very important result about induced representations is the Frobenius reciprocity theorem which connects the operations Ind and Res.

Theorem 5.10.1 (Frobenius reciprocity)**.** *Let $H \subset G$ be groups, V a representation of G and W a representation of H. Then the space* $\mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$ *is naturally isomorphic to* $\mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$.

Proof. Let $E = \mathrm{Hom}_G(V, \mathrm{Ind}_H^G W)$ and $E' = \mathrm{Hom}_H(\mathrm{Res}_H^G V, W)$. Define $F : E \to E'$ and $F' : E' \to E$ as follows: $F(\alpha)v = (\alpha v)(e)$ for any $\alpha \in E$ and $(F'(\beta)v)(x) = \beta(xv)$ for any $\beta \in E'$.

In order to check that F and F' are well defined and inverse to each other, we need to check the following five statements.

Let $\alpha \in E$, $\beta \in E'$, $v \in V$, and $x, g \in G$.

(a) $F(\alpha)$ is an H-homomorphism; i.e., $F(\alpha)hv = hF(\alpha)v$.

Indeed, $F(\alpha)hv = (\alpha hv)(e) = (h\alpha v)(e) = (\alpha v)(he) = (\alpha v)(eh) = h \cdot (\alpha v)(e) = hF(\alpha)v$.

(b) $F'(\beta)v \in \mathrm{Ind}_H^G W$; i.e., $(F'(\beta)v)(hx) = h(F'(\beta)v)(x)$.

Indeed, $(F'(\beta)v)(hx) = \beta(hxv) = h\beta(xv) = h(F'(\beta)v)(x)$.

(c) $F'(\beta)$ is a G-homomorphism; i.e. $F'(\beta)gv = g(F'(\beta)v)$.

Indeed, $(F'(\beta)gv)(x) = \beta(xgv) = (F'(\beta)v)(xg) = (g(F'(\beta)v))(x)$.

(d) $F \circ F' = \mathrm{Id}_{E'}$.

This holds since $F(F'(\beta))v = (F'(\beta)v)(e) = \beta(v)$.

(e) $F' \circ F = \mathrm{Id}_E$; i.e., $(F'(F(\alpha))v)(x) = (\alpha v)(x)$.

Indeed, $(F'(F(\alpha))v)(x) = F(\alpha xv) = (\alpha xv)(e) = (x\alpha v)(e) = (\alpha v)(x)$, and we are done. □

Problem 5.10.2. The purpose of this problem is to understand the notions of restricted and induced representations as part of a more advanced framework. This framework is the notion of tensor products over k-algebras. In particular, this understanding will lead us to a